Lösungen · Trigonometrie

1. $\boldsymbol{\varepsilon = \cos^{-1}(\frac{13}{32}) \approx 66{,}0°}$ ... $\cos\varepsilon = \frac{13}{32}$

2. $\boldsymbol{\beta_1 = \cos^{-1}(\frac{255}{384}) \approx 48{,}4°}$ ... $\cos\beta_1 = \frac{255}{384}$

3. $\boldsymbol{\alpha = \sin^{-1}(\frac{7{,}4}{14{,}1}) \approx 31{,}7°}$ ... $\sin\alpha = \frac{7{,}4}{14{,}1}$

4. $\boldsymbol{\alpha = \cos^{-1}(\frac{4}{9}) \approx 63{,}6°}$ ... $\cos\alpha = \frac{4}{9}$

5. $\boldsymbol{\alpha = \tan^{-1}(\frac{14}{9}) \approx 57{,}3°}$ ... $\tan\alpha = \frac{14}{9}$

6. $\boldsymbol{x = 2\sqrt{7289} \approx 170{,}8\text{ cm};\ \alpha = \tan^{-1}(\frac{16}{170}) \approx 5{,}4°}$ ...
$x^2 = 170^2 + 16^2$; $\tan\alpha = \frac{16}{170}$

7. $\boldsymbol{BC = \sqrt{20} \approx 4{,}47;\ \beta = \tan^{-1}(2) \approx 63{,}4°}$ ... $BC^2 = 2^2 + 4^2$; $\tan\beta = \frac{4}{2}$

8. $\boldsymbol{a = \frac{7{,}4}{\sin 52°} \approx 9{,}4\text{ m}}$ ... $\sin 52° = \frac{7{,}4}{a}$

9. **a)** $\boldsymbol{AB \approx 1228{,}7\text{ m}}$ ... $\sin 55° = AB : 1500$

b) $\boldsymbol{AD \approx 860{,}4\text{ m}}$ ... $\cos 55° = AD : 1500$

10. $\boldsymbol{PB = \frac{1{,}20}{\tan 34{,}9°} \approx 1{,}72\text{ m}}$ ... $\tan 34{,}9° = 1{,}20 : PB$

11. $\boldsymbol{AC = \frac{h}{\sin 22°} \approx 78{,}1\text{ cm}}$ ... $h = \sqrt{32^2 - 13^2} = 3\sqrt{95}$; $\sin 22° = h : AC$

12. $\boldsymbol{BF \approx 93{,}0\text{ cm}}$ ... $\sin 45° = BF : 131{,}5$
$\boldsymbol{BC = BF : \cos 65° \approx 220{,}0\text{ cm}}$ ... $\cos 65° = BF : BC$

13. $\boldsymbol{\text{Berg} = 57{,}9\text{ m};\ D = 69{,}9\text{ m}}$ ... $\sin 30° = BC : 112{,}8$; $\text{Berg} = BC + 1{,}5$; $D = 57{,}9 + 12{,}0$

14. $\boldsymbol{BD \approx 502\text{ m}}$ ... $\sin 54° = AD : 920$; $\tan 34° = BD : AD$

15. $\boldsymbol{AC = 12{,}0 \cdot \frac{\sin 55°}{\sin 5°} \approx 112{,}8\text{ m}}$ ... $\gamma = 35° - 30°$; $AC : \sin 55° = \frac{12{,}0}{\sin 5°}$

16. $\boldsymbol{BD = 4{,}1 \cdot \frac{\sin 123°}{\sin 36°} \approx 5{,}85\text{ cm}}$ ... $BD : \sin 123° = \frac{4{,}1}{\sin 36°}$

17. $\boldsymbol{y = 160 \cdot \frac{\sin 141°}{\sin 34°} \approx 180{,}1\text{ cm}}$ ... $\frac{y}{\sin 141°} = \frac{160}{\sin 34°}$

18. $\boldsymbol{BC = 384 \cdot \frac{\sin 38°}{\sin 34°} \approx 422{,}8\text{ m}}$ ... $BC : \sin 38° = \frac{384}{\sin 34°}$

19. $\boldsymbol{BC = 4{,}50 \cdot \frac{\sin 62{,}8°}{\sin 34{,}9°} \approx 7{,}00\text{ m};\ A \approx 92{,}0\text{ m}^2}$ ...
$BC : \sin 62{,}8° = \frac{4{,}50}{\sin 34{,}9°}$; $A = 8{,}00 \cdot (4{,}50 + 7{,}00)$

20. $\boldsymbol{\text{Weg} \approx 5665\text{ m}}$ ... $JR : \sin 60° = \frac{2800}{\sin 50°}$; $\text{Weg} = JR + 2500$

21. **nein** ... $\angle DBC = 109° - 35° = 74°$; $BC : \sin 60° = \frac{2004}{\sin 74°}$; $BC = 2004 \cdot \frac{\sin 60°}{\sin 74°} \approx 1805{,}5$ m

22. $\boldsymbol{DP = 11 \cdot \frac{\sin 115°}{\sin 49°} - 10 \approx 3{,}2\text{ m}}$ ... $DE : \sin 115° = \frac{11}{\sin 49°}$; $DP = DE - 10$

23. $\boldsymbol{AD = 9 \cdot \frac{\sin 60°}{\sin 80°} \approx 7{,}9\text{ m}}$ ... $\angle ACD = 86° - 26° = 60°$; $AD : \sin 60° = \frac{9}{\sin 80°}$